

What am I building?

Webtraffic Counter

- Count Webtraffic using a 7-segment display, it should accurately update the amount of visitors to our ChyPhy website in real time
- It should be clean and cute in the office

Goal/Hypothesis:

What are you testing or trying to achieve?

Webby the website counter for our office!

Steps:

List 5–8 bullets describing the actions you'll take.

- **Learn 7 segment display logic via wokwi**
- **Learn how to reduce wires on wokwi**
- **Learn how to make a good algorithm that aligns with our web counter without an outside api (ideate an algorithm that mimics that behavior)**
- **Build Webby in real life!**
- **Make webby work with the website counter**
- **Make webby a pretty home**
- **Give Webby a voice!**

Timebox:

Set a strict duration (e.g., 90 minutes) for the experiment.

DECEMBER 1ST

Problem Statement:

Describe what you're trying to show, build, or test in one or two lines.

Build a working 7-segment display that utilizes code to visually track traffic to our lab's website

Success Criteria: Define clear, measurable outcomes that tell you when you're done (e.g., accuracy $\geq 80\%$, working prototype that ingests sample data).

1. **Wokwi: Webby Works by displaying numbers from 0-999,999 from an algorithm**
2. **Build it in real life: Webby works in real life!**
3. **Website counter: Webby displays website traffic within 10 seconds of a visitor**
4. **Make it Cute: Webby makes a quiet noise or gesture inside a cute home**

Final output:

- Working and cute looking WebCounter (Webby!)

Deliverables Ladder: List a Bronze (bare minimum), Silver (usable), and Gold (nice-to-have) version of your deliverable.

Bronze:

- Webby can display numbers accurate to our traffic data

Silver:

- Webby can display numbers accurate to our traffic data, with minimal hardware and a clean design

Gold:

- Silver but with a cute home for Webby to live in
- Also able to be chargeable so it can be placed anywhere and last a long time without being charged
 - o Display when battery low?
- Has a cute noise or display when we get a visitor or a celebration when there are a lot of visitors

Constraints:

Note any limitations in data, compute resources, libraries, or time.

1. Time

- a. I want this to be finished by the END OF NOVEMBER

2. Resources

- a. Resources exist (wires, arduino, etc.) but are accessed in the lab so I need to consider time management and getting it done in a timely manner at the lab (can't do this all at home)
 - i. Also, can order stuff with a need basis by discussing with supervisor (Carlos)
- b. We have the 3D printer but it could take a while to learn how to use it accurately, it would be nice to brush off the dust on it though

3. Data

- a. I am very unknowledgeable in this topic so I need to do a lot more research than others in this lab, and the unknowns are intimidating making me take too much time researching the wrong things instead of focusing on what should be done

Known Unknowns:

Capture the questions or uncertainties that need research.

- How can I connect the wires to accurately light up each segment?
- How can I implement buttons that go up and down correctly for the numbers in the code
- How does it look when I implement more than one display? Is it more complicated? How many more wires do I need?
- IS THERE A DEVICE THAT CAN REDUCE THE AMOUNT OF WIRES WEBBY NEEDS TO WORK WHILE PROVIDING THE SAME AMOUNT OF POWER BUT IN A SMALLER SETTING?
- How can Webby connect to the website?
 - o How long does it take to update a visitor?
- How can I make Webby's 3D home?
- How can I make Webby display a noise or another message for visitor traffic and how can I connect it accurately and without taking up too much space?
- How much power does a 7-segment display use and how could I get it to last for at least a week without needing to charge?
- How can I get this project done only focusing on what I need and mitigating time I spend on useless things (I know this is unavoidable but I'm going a little far with the research and no results thing)

Milestones & Dates:

Set 3–5 checkpoints with target dates to track progress.

1. BY Monday Nov. 3,
 - a. From 26-3: Webby correctly displays numbers and can move to real life practice
2. Monday Nov. 10,
 - a. From 3-10: Webby Works in real life and I at least have an idea of how to lessen the wires and make the hardware cleaner
3. Monday Nov. 17,
 - a. From 10-17: Webby can accurately count website traffic, or maybe at least I have a good idea of how to connect him
4. Monday Nov. 24,
 - a. From 17-24: Webby Iteration (less wires, webby works with traffic)
5. Monday Dec. 1,
 - a. From 24-1: Webby's home has been built and we have put him in someplace nice and cozy

Review Cadence:

Decide how often you'll sync with collaborators (e.g., weekly)

- Every Monday: Meeting with Lab
- Every Wednesday: In person work before classes
- More?

Plan Small Experiments

Break your project into tiny experiment tickets that you can finish in two hours or less. Each ticket should have a clear goal and steps.

1. Work with Webby on Wokwi

- a. Webby (7seg display), displays each segment correctly (in order from a-g) corresponding to the code created
- b. Webby can display a number from 0 to 9
- c. Webby can display a number from 0-99
- d. Webby can display a number from 0-999,999
- e. Webby can iterate through the numbers going up and down using buttons
 - i. Case switches here?
- f. Webby can iterate through numbers based on input
 - i. (Maybe not yet, have to understand website counter logic? Let's think this through a little more)

2. Webby is real!

- a. Build the **final** (see final version from step 1) wokwi version on an actual Arduino and see it light up in real life!
- b. Figure out how to iterate it to have less wires and use a smaller device to make it work (**Help!!**)

3. Webby can count traffic :3

- a. Talk to Carlos about how to implement this, but it needs to connect with webby
- b. Webby needs to be able to take input from our website, and make a decision on how to display it with a consistent power source in our office
 - i. Is he chargeable or need to be plugged in all the time?

4. Webby's new home

- a. If we got time I want to 3d print a model that takes into account all wiring and hardware and put webby in and make him pretty for the office
- b. Add a non-annoying noise or visual cue (if a lot at one time, he waits until the traffic slows down and is able to keep track of the number while displaying a cute message)

Metrics & Exit Criteria:

Decide how you'll measure success and when to stop or pivot.

Success:

- When 25 minutes produces either information to move forward or the desired output
- When I have achieved the goals set in milestones and finished an experi

Stop:

- When I've been doing nothing for 25 minutes and feel stuck

Pivot:

- When I feel stuck (in stop phase) I do 10 minutes of research either asking for help from a person, youtube, or chatgpt!

Tracking Progress:

Record the outcome and outline what you will do next

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What I did:

What I learned:

What I will do next: